Feature Engineering & Nonlinearity

QLS–MiCM Workshop

Why This Matters

Real biological and clinical effects are **rarely linear**.

Examples:

- dose–response curves.
- saturation and thresholds.
- diminishing returns.
- nonlinear gene/protein interactions.

Key Idea: Add Nonlinear Features

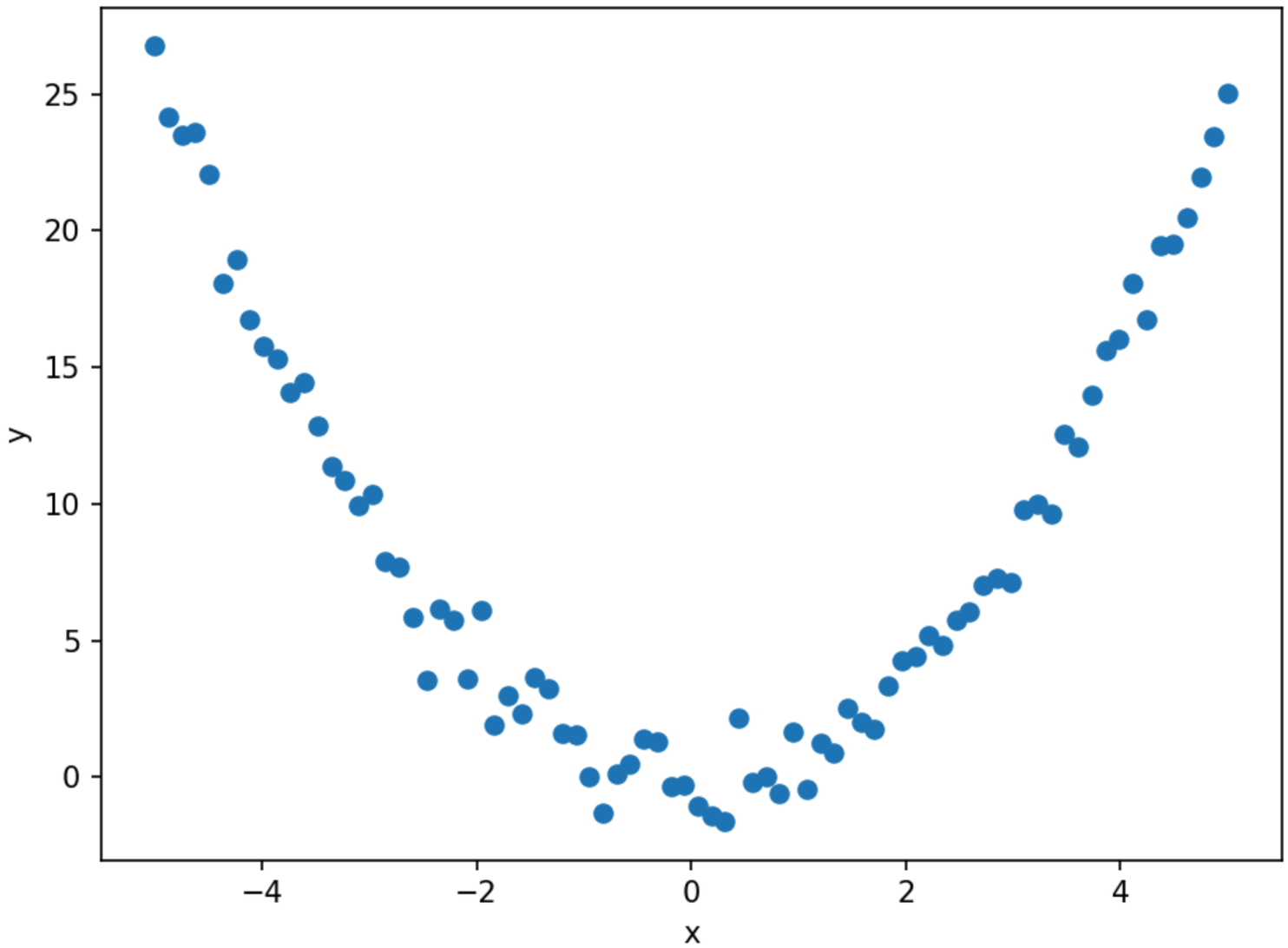
We keep using a **linear model**, but expand the input features:

Polynomial Features

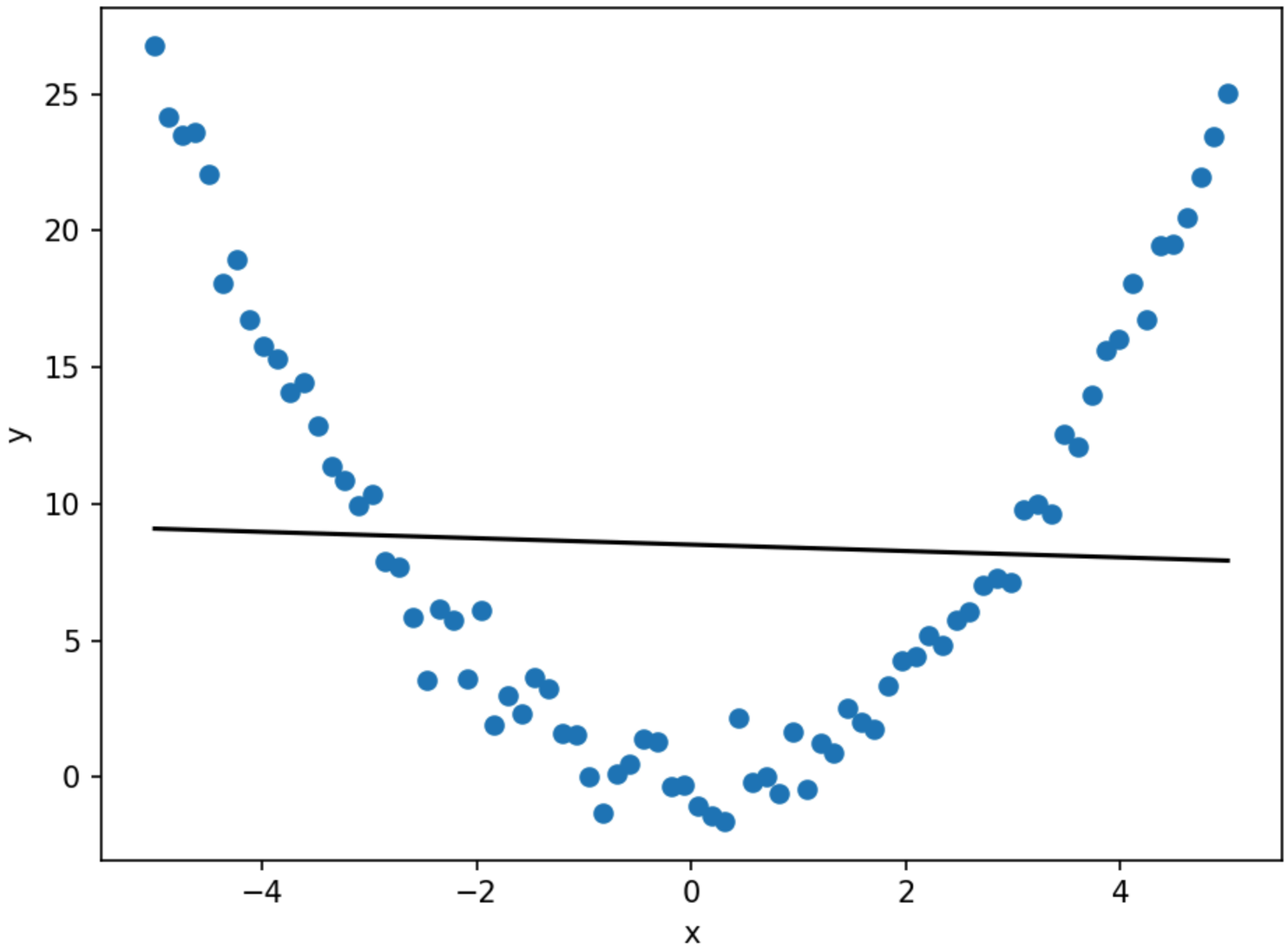
- x^2, x^3.
- interactions ($x_1 x_1, x_1^2, x_2, \ldots$).
- allows curvature and richer patterns.

This is still simple, interpretable, fast.

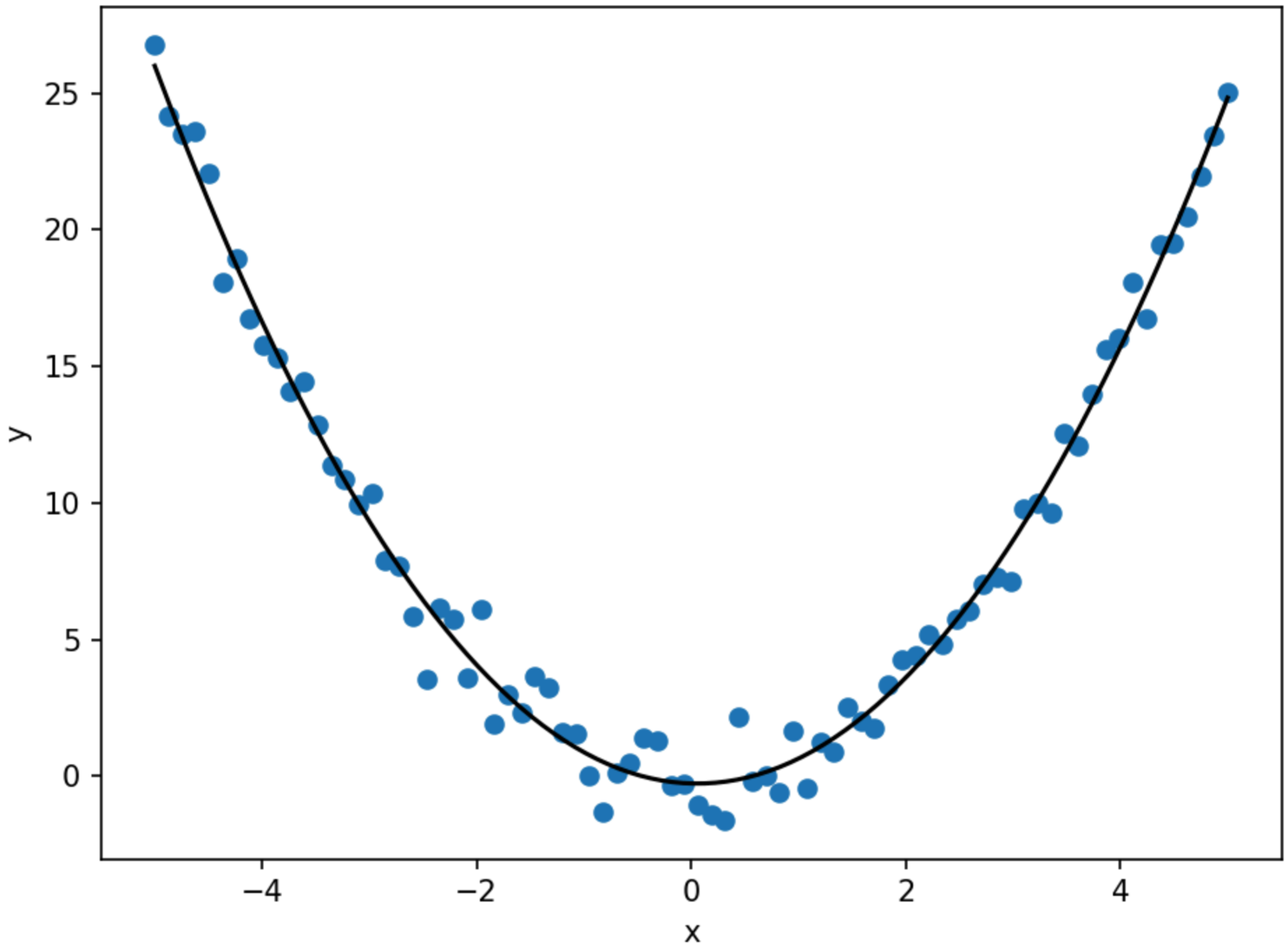
Parabolic Relationship (Point Cloud)



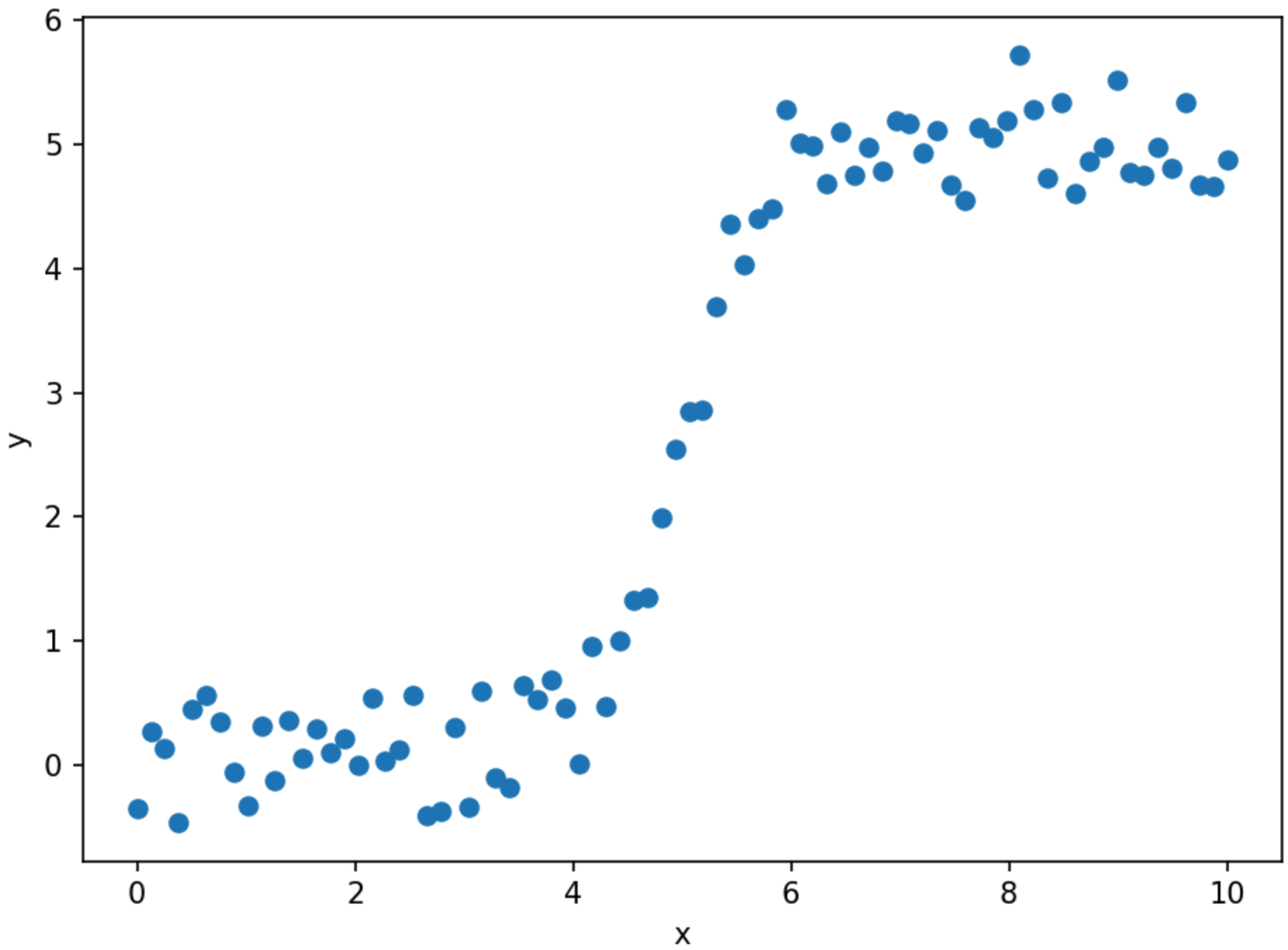
Linear Fit on Parabolic Data (Underfitting)



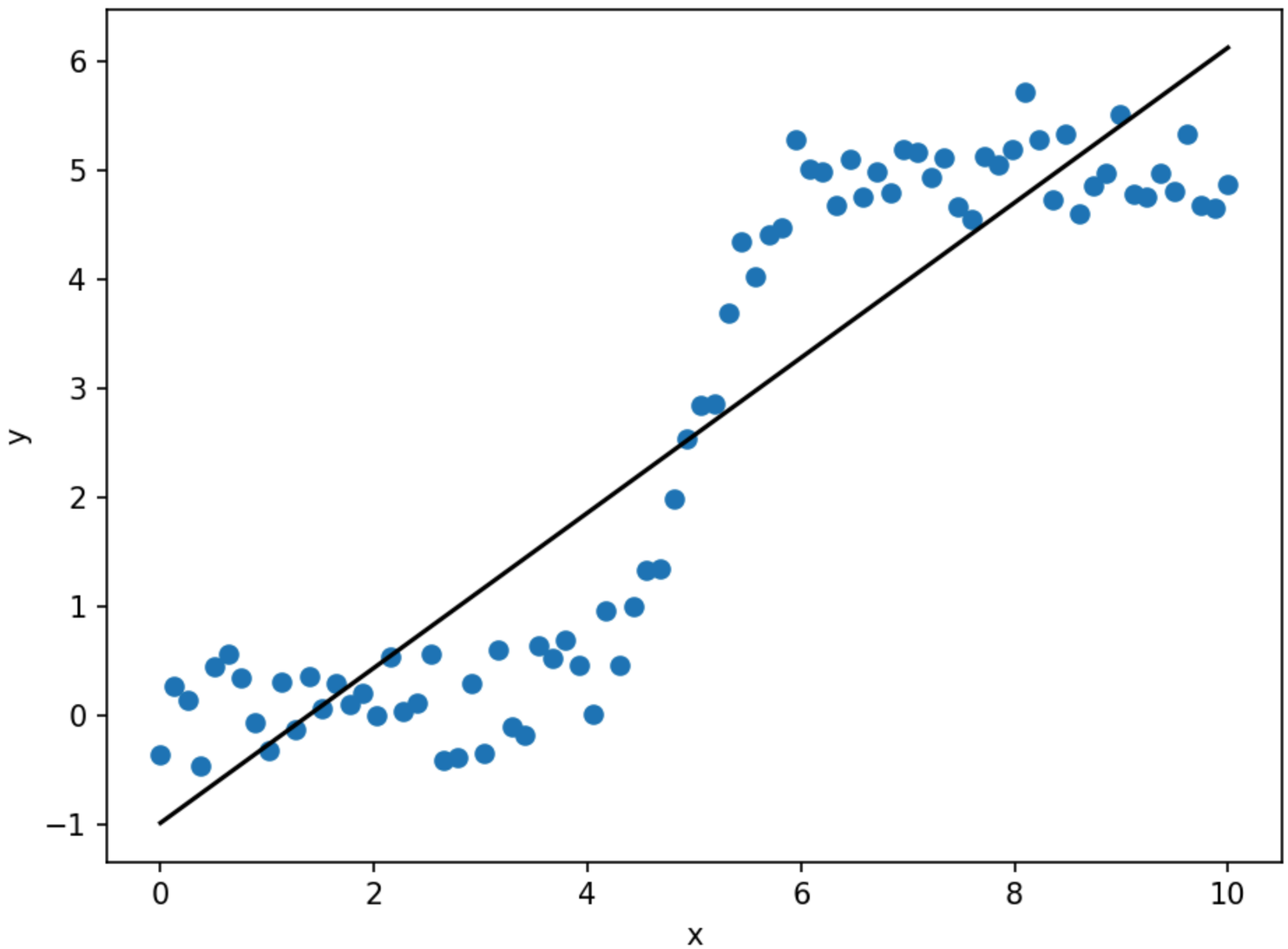
Quadratic Fit on Parabolic Data (Good Fit)



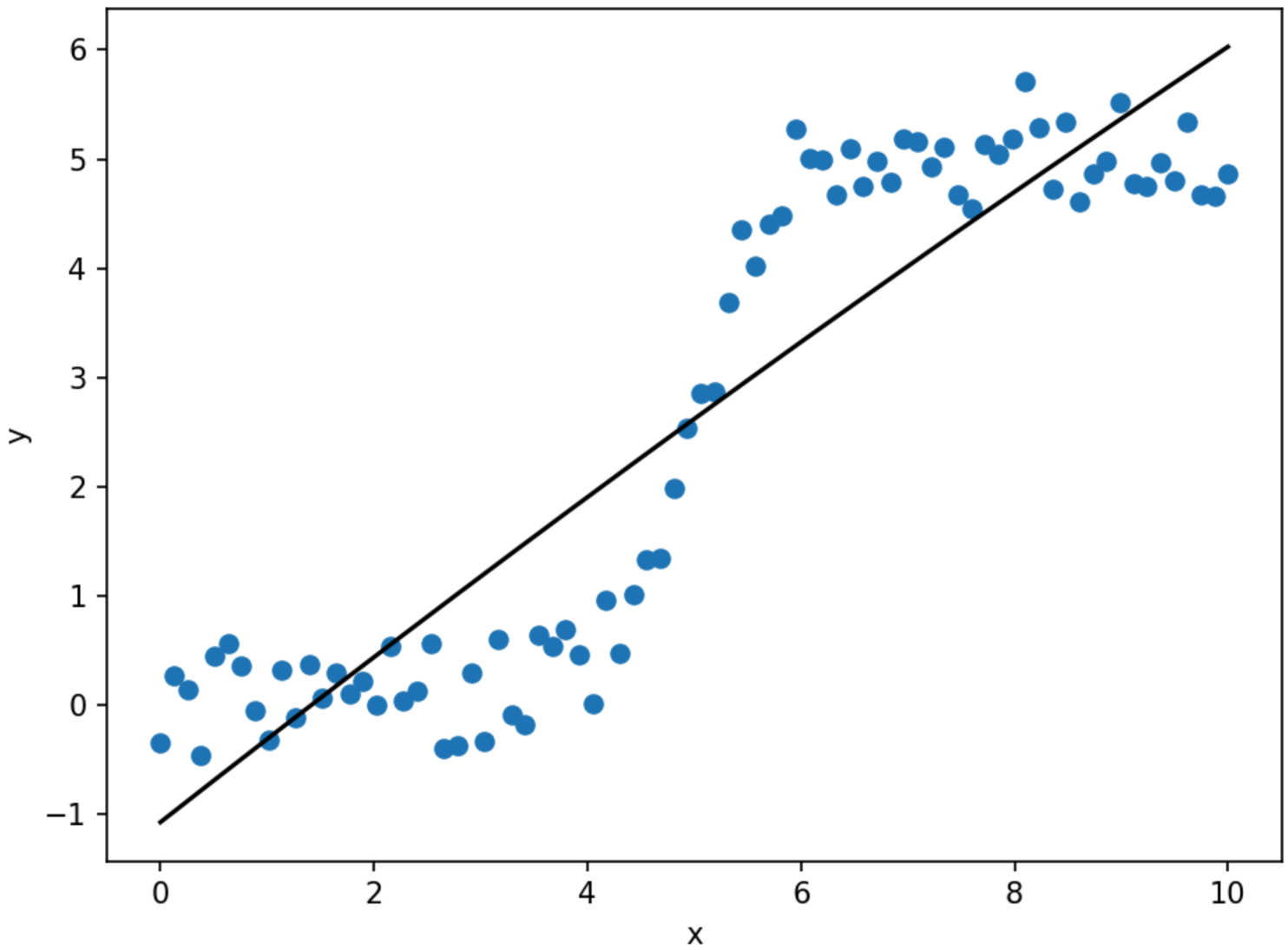
S-Shaped Dose–Response (Point Cloud)



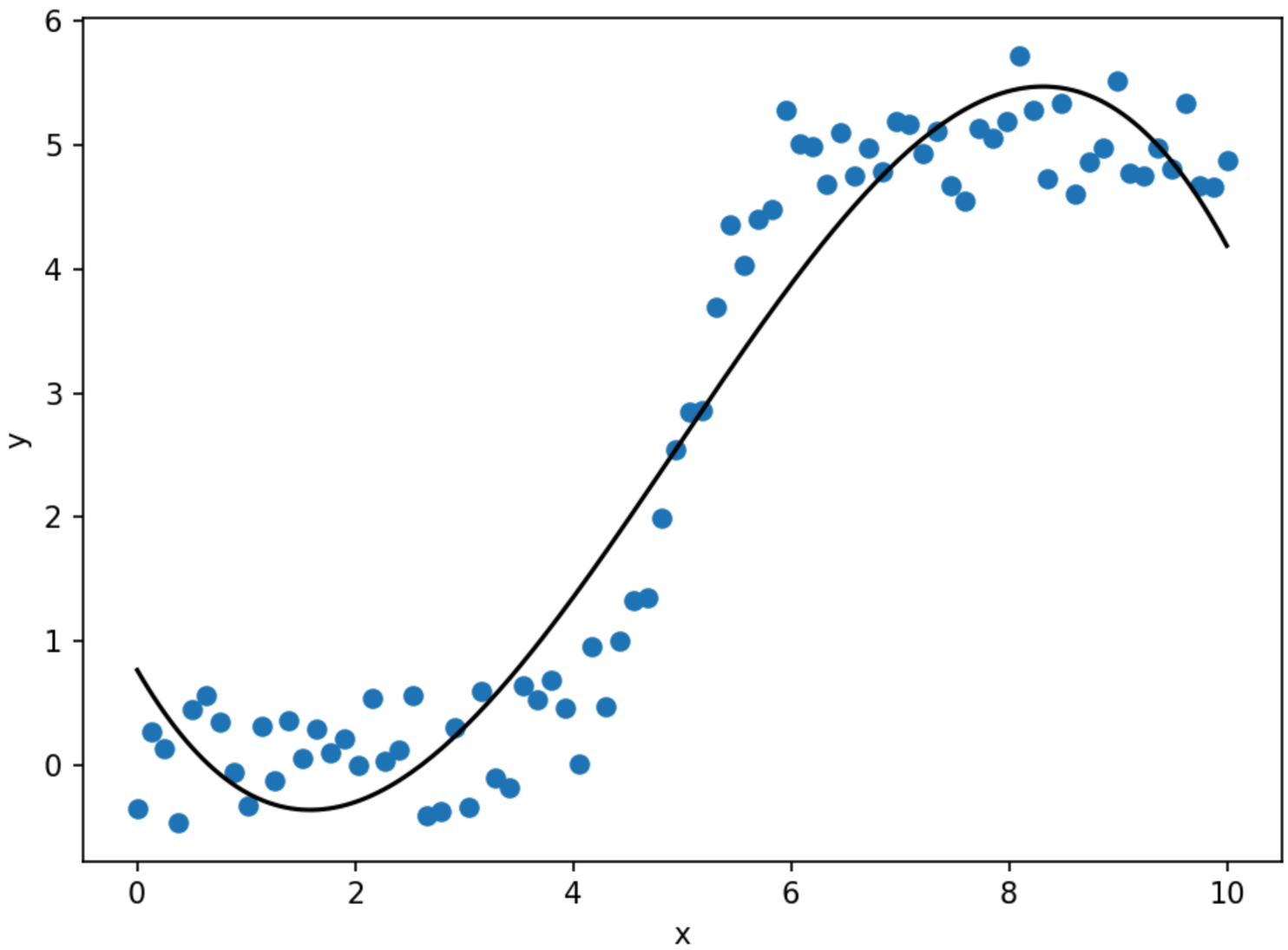
Linear Fit on S-Shaped Dose–Response (Poor Fit)



Quadratic Fit on S-Shaped Dose–Response (Closer Fit)



Cubic Fit on S-Shaped Dose–Response (Best Polynomial Fit)



For Further Learning

Sparse Identification of nonlinear dynamical systems:

- Discovered dynamics of fluid vortex shedding behind an obstacle
- Input many nonlinear features to the model, used L1 to select the relevant features.
- https://arxiv.org/abs/1509.03580

Spline Regression:

- More flexible non-linear regression to fit complex curves

Workflow You'll Use

1. Load the drug response dataset.
2. Fit a **baseline linear model**.
3. Add polynomial features (degree=???).
4. Compare model performance.
5. Visualize:
   - data vs linear fit.
   - data vs polynomial fit.

What to Look For

- Do non-linear functions of x improve MAE?
- Does the polynomial model better follow the data trend?
- Is the relationship between top predictor + effect nonlinear?
- When does added flexibility risk overfitting?